

Scalability & Future Plan

Date	15 March 2024
Team ID	SWTID-2026-5650
Project Name	ComicCraft - AI Comic Story Creator using Gemini Models
Maximum Marks	1 Mark

Scalability & Future Plan:

This document outlines how the current project solution can be scaled to handle larger user bases, increased data loads, or extended features in the future. It also captures the team's roadmap for enhancing and evolving the project beyond its current state.

Current System Limitations:

S.No	Limitation	Impact	Priority to Address (High / Medium / Low)
1	Gemini API Rate Limits & Token Overhead	Causes latency during peak concurrent comic generation requests.	High
2	Local Storage of High-Res Comic Images	Limits scalability of user story archives on standard web server storage.	Medium
3	Character Visual Consistency Across 10+ Panels	Complex prompt drift occurs when story arcs exceed 8-10 consecutive panels.	High

Scalability Plan:

S.No	Scalability Aspect	Current State	Proposed Upgrade / Solution
1	User Load	Single-instance server handling up to 50 concurrent users	Implement distributed microservices with Redis caching and AWS Auto Scaling groups.
2	Data Storage	Local disk storage for comic images and metadata	Migrate image asset storage to AWS S3 / Cloudflare R2 with CloudFront CDN distribution.
3	Performance	Synchronous Gemini API processing (~8s per panel generation)	Transition to asynchronous Celery background workers with WebSockets for real-time updates.
4	Security	Basic API key authentication & environment variables	Deploy OAuth2 / JWT user authentication, API rate limiting, and KMS secret management.

Future Roadmap:

Phase	Planned Feature / Enhancement	Target Timeline	Expected Impact
Phase 1	Core Gemini Comic Generator, Panel Layouts & PDF Export	Month 1 - 3	Functional MVP with end-to-end comic creation capabilities.
Phase 2	Multi-lingual Speech & Dynamic Voiceover Synthesis per Character	Month 4 - 6	Enhanced accessibility and global reach for comic creators.
Phase 3	Fine-tuned Custom Gemini LoRA Models for Specific Art Styles	Month 7 - 9	Superior visual artistic consistency and user creative control.
Phase 4	Interactive Web-based Collaborative Comic Creator & Publishing Platform	Month 10 - 12	Enterprise & community ecosystem with monetization options.